

CPE 241 Database Management System

Problem Session #08

Date: check with PS08 in LEB2

Peer Feedback and subquery

Introduction

In this lab, you will gather feedback from two other groups working on different projects. This will help ensure that your database design is clear, well-structured, and understandable to those who were not involved in its development. Effective communication of your design is essential in real-world database projects, where teams must collaborate and validate each other's work. Beyond collecting feedback, you will also design and implement three SQL queries that use subqueries. These queries should demonstrate complex data retrieval techniques, such as filtering results, performing calculations, or aggregating data based on nested conditions. This will enhance your ability to write efficient and structured SQL queries while reinforcing your understanding of relational database operations.

By the end of this lab, you will have strengthened both your database communication skills and SQL proficiency in handling advanced queries.

Objectives

By completing this lab, you will:

1. Effectively Express the Database and UX/UI Design

- Present both **database structure and user interface design** to groups working on different projects.
- Ensure that the design is **clear, logical, and understandable** to those unfamiliar with it.
- Explain how **user experience (UX)** and **user interface (UI)** decisions align with the database structure and functionality.

2. Gather and Apply Constructive Feedback

- Identify **strengths and weaknesses** in both the **database** and **UX/UI design** through external review.
- Use feedback to **refine and improve** the system's usability and data organization.

3. Develop Advanced SQL Queries Using Subqueries

- Design **three SQL queries** that utilize **subqueries** for advanced data retrieval.
- Implement queries for use cases such as **nested filtering, aggregation, or conditional selections**.
- Optimize queries for **performance and efficiency**.

4. Justify UX/UI and Query Design Choices

- Explain **why certain UX/UI decisions** were made and how they improve usability.
 - Describe **why subqueries were used** in each SQL problem and discuss **performance trade-offs**.
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Steps to Complete the Lab

Step 1: Plan and Design Subqueries

- Before presenting their work, students should **analyze their database** and identify **three meaningful use cases** where subqueries would be useful.
- **Examples of subquery use cases:**
 - **Filtering results based on conditions from another query**
 - **Aggregating data before using it in a main query**
 - **Comparing values between tables dynamically**
- Write and test **three SQL queries** using subqueries.

Step 2: Prepare for the Presentation

- Students should **review their database and UX/UI design** to ensure it is clear and well-structured.
- Identify key points they will **explain to another group**, including:
 - How the **database is structured** and why it was designed this way.
 - How **UX/UI choices** relate to the database.
 - **The purpose of their subqueries** and how they solve real-world data problems.
- Create a **brief summary or slide** to support their explanation (optional but useful).

Step 3: Present to Two Other Groups

- Students will **pair up with two other groups working on different projects**.
- Each group takes turns **presenting** their design and UX/UI decisions.
- While presenting, they should:
 - **Explain their database and user interface** clearly.
 - **Describe their three subqueries** and how they retrieve useful data.
 - Answer any **questions or challenges** raised by the other groups.

Step 4: Gather Feedback and Discuss Improvements

- The **listening group** should provide feedback on:
 - **Clarity** of the database structure and UX/UI design.
 - **Potential issues** in data relationships, UI navigation, or database logic.
 - **Effectiveness of subqueries** in solving data retrieval problems.
- Each group should **note down suggestions** and discuss how to refine their work.

Step 5: Refine and Submit Work

- After receiving feedback, students should:
 - Make **adjustments** to their **database structure or UX/UI if needed**.
 - **Optimize or refine subqueries** based on feedback.
 - Submit the following:
 - **Refined database and UX/UI design** (if changes were made).
 - **Three final subqueries** with explanations of their purpose and function.
 - **A brief summary of the feedback received and what was improved**.
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Deliverables

1. **Three Finalized SQL Subqueries**
 - Each subquery should be well-documented with an explanation of its **purpose** and **functionality**.
2. **Refined Database and UX/UI Design** *(if modifications were made based on feedback)*
 - Updated **ERD** or **schema adjustments**.
 - Revised **UI wireframes** or **user flow diagrams**.
3. **Presentation Summary**
 - A brief **overview** of their database and UX/UI design.
 - **Key insights** from presenting to other groups.
4. **Feedback and Improvements Report**
 - **Summary of feedback** received from peer groups.
 - **List of changes** made based on that feedback.

Submission

Combine results from all deliverables in a PDF file then submit the report to PS08 on LEB2 before the deadline.